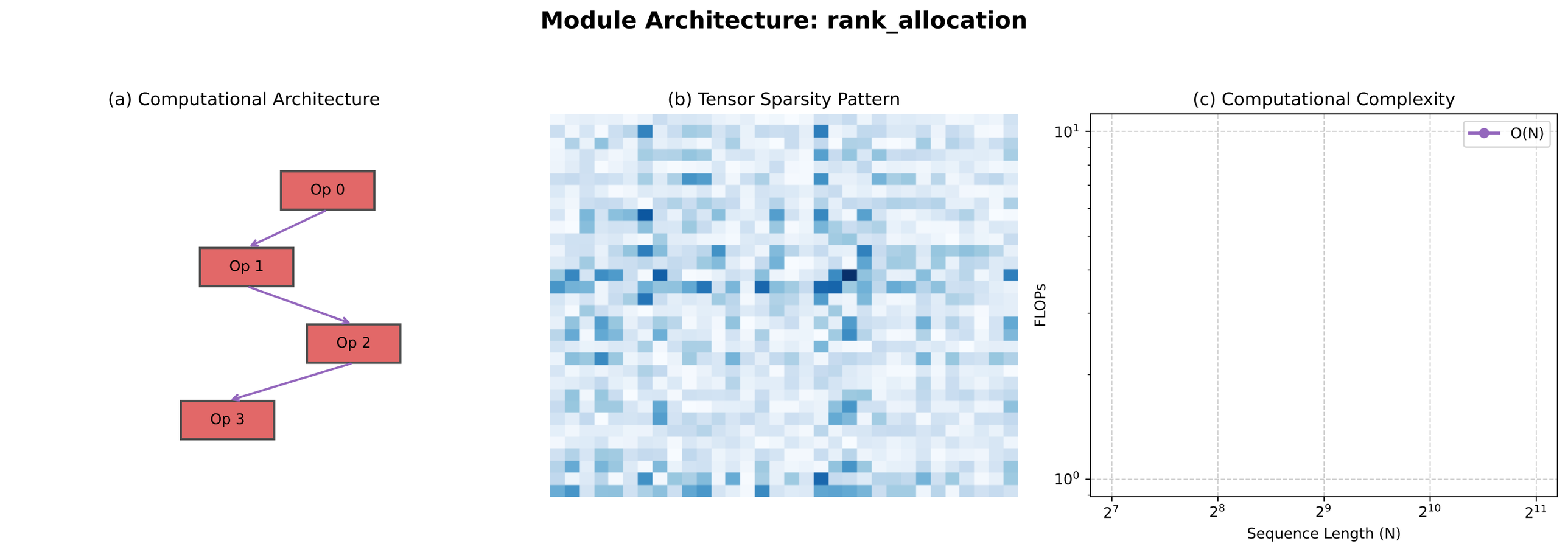Module Architecture: rank_allocation